

Akshansh Dev Guide — Complete Reference

Private document — everything built, every issue faced, how to maintain it all

Last updated: June 17, 2026 | **Author:** Akshansh Yadav | **Scope:** All 7 projects + GitHub profile + resume PDF + deployment history

0. Complete Tech Stack — Everything Used

This section is a single-glance reference for every technology, library, framework, service, and protocol used or actively learned across all projects.

Category	Tools / Technologies	Where Used
Languages	TypeScript, JavaScript (ES2024), Python 3.12, C, C++ (C++17), SQL, Solidity, Rust (learning), Bash	SwapX (TS/JS), FaceMetrics (JS/Py), RC Car (Python/C), Tic-Tac-Toe (JS/Py/C), Space Shooter (JS), Trading Sim (Py)
Frontend Frameworks	Next.js 14 (App Router, SSR, API Routes), React 18 (hooks, context, suspense), Tailwind CSS (utility-first)	SwapX
Runtime & Backend	Node.js 22, Express.js (familiarity), Next.js API Routes (serverless functions)	SwapX API layer
Auth & Security	NextAuth.js v4 (Google OAuth provider), OAuth 2.0 flow, JWT session tokens, NEXTAUTH_SECRET (256-bit random)	SwapX login system
Validation	Zod (runtime schema validation + TypeScript inference)	SwapX API input validation
Database / Cache	PostgreSQL (relational DB, ACID, parameterised queries, studied across all guides), Redis (data structures: Hashes, Lists, Strings), Upstash Redis (serverless REST API, not TCP — works in Edge/Serverless)	SwapX portfolio, trade history, 60s price cache
External APIs	CoinGecko API (free tier, 50 req/min, no API key needed), Google Cloud OAuth API, Upstash REST API	SwapX real-time prices + auth
UI Libraries	Recharts (composable chart library), Lucide React (icon set), clsx (conditional classNames), shadcn/ui (component patterns)	SwapX charts, icons, buttons
Web APIs	Canvas API (2D rendering context), Web Audio API (game sounds), WebSockets (learning), Fetch API, getUserMedia (webcam)	Space Shooter, FaceMetrics webcam, SwapX fetch
Computer Vision	OpenCV (image processing, brightness, contours), MediaPipe FaceMesh (468 facial landmarks, browser WASM), NumPy (array math)	FaceMetrics (all), RC Car gesture control
Speech Recognition	Vosk (offline speech-to-text, English + Hindi models, runs on-device — no cloud)	RC Car voice control
Data & Scientific	pandas (DataFrames), NumPy, scikit-learn (ML algorithms: linear regression, SVM, k-NN, decision trees), yfinance (Yahoo Finance stock data), matplotlib (plotting)	Trading Strategy Simulator, FaceMetrics
Python GUI / Game	Tkinter (desktop GUI), Kivy (cross-platform UI), pygame (2D game engine)	FaceMetrics v1–v15, Space Shooter (original Python version)
Embedded Systems	Arduino / ESP32 (microcontrollers), Embedded C, L298N motor driver, HC-SR04 ultrasonic sensor, LiPo battery management, UART serial, UDP sockets, PWM signals	Gesture+Voice RC Car
Networking & Protocols	UDP sockets (Python ↔ ESP32 real-time command relay), REST (JSON over HTTP), OAuth 2.0 redirect flow	RC Car, SwapX
Hosting & Deploy	GitHub Pages (static sites, free, auto-deploy on push), Railway (Node.js server, free tier 750h/mo, sleeps after 15 min idle)	All GitHub Pages sites, SwapX on Railway
DevOps / CI-CD	Git (version control), GitHub (remote + Pages + profile README), GitHub Actions (learning), Docker (learning), Linux / Ubuntu (server environments, WSL, container base images, bash scripting)	All projects
TypeScript Ecosystem	@types/node, @types/react, @types/gapi, @types/gapi.auth2, ESLint + eslint-config-next, tsconfig path aliases	SwapX type safety

Web3 (Exploring)	Solana (SPL tokens, Phantom wallet), Ethereum (EVM, gas), Solidity (smart contracts), ethers.js (EVM interactions), Anchor framework (Solana programs), DeFi mechanics (AMM, order-book, liquidity pools, slippage)	SwapX DEX logic (simulated), active self-study
AI & LLMs (Exploring)	PyTorch (tensors, autograd, training loop), HuggingFace Transformers (AutoTokenizer, AutoModel, pipeline API), LangChain (chains, agents, ReAct loop), FastAPI (ML model serving), ChromaDB (vector DB, embeddings), Ollama (local LLM inference), RAG pipelines (chunk → embed → retrieve → generate)	Active self-study; NeuraCore guide project
Cloud Services (Manual)	Google Cloud Console (OAuth 2.0 credentials, consent screen config), Upstash (managed Redis, REST endpoint + token), Railway (build + deploy + env vars + logs)	SwapX production
Package Managers	npm v11 (Node.js packages), pip (Python packages), winget (Windows package manager for Node.js install)	All projects
Tooling	VS Code, PowerShell 5.1, Chrome Headless (PDF generation), Nixpacks (Railway build system), railway CLI v5, gh CLI	Development + deployment workflow

1. Projects Overview

Project	Tech	Live URL	Local Path	Hosting
Portfolio / Resume	HTML/CSS/JS	expediator.github.io/resume	Desktop\Akshansh-Resume\	GitHub Pages
SwapX (CEX+DEX)	Next.js 14 · TypeScript · Redis	railway.app/...	Desktop\repos\swapx-dex\	Railway (free)
FaceMetrics	JS · MediaPipe · Canvas	expediator.github.io/face-rater	Desktop\repos\face-rater\	GitHub Pages
Space Shooter	JS · Canvas API	expediator.github.io/space-shooter	Desktop\repos\space-shooter\	GitHub Pages
Tic-Tac-Toe AI	JS · Minimax · C · Python	expediator.github.io/tic-tac-toe	Desktop\repos\tic-tac-toe\	GitHub Pages
Gesture+Voice RC Car	Python · ESP32 · MediaPipe · Vosk	—	Desktop\repos\gesture-controlled-robot\	—
Trading Sim	Python · pandas · yfinance	—	Desktop\repos\trading-strategy-simulator\	—

2. Complete Session History — What Was Built & Why

2.1 Space Shooter — Rewrite from Python to JS Canvas

Original Space Shooter used **pygbag** (Python-to-browser compiler). It failed to load in browsers — pygbag produced a Python WASM bundle that required specific server headers and had long load times.

Decision: Rewrote the entire game in pure JavaScript using the HTML5 Canvas API. No frameworks, no build step — just a single `index.html` file that loads instantly.

Features built: thrust+inertia physics (WASD/arrows), two AI enemy types (seeker + wandering boss), power-ups (Rapid Fire, Shield), particle explosion system, full HUD, touch/mobile controls.

2.2 FaceMetrics — Webcam Integration

FaceMetrics previously only allowed static image uploads. Added live webcam capture so users can take a photo directly without needing to save/upload a file first. Also added auto-brightness enhancement so dark/underexposed photos score correctly.

2.3 FaceMetrics — Rename from "Face Rater"

Original repo and project was called "Face Rater." Renamed to **FaceMetrics** across the resume website, resume PDF, and GitHub descriptions to sound more professional for job applications.

2.4 Trading Strategy Simulator — Removed from Resume

Removed from resume and portfolio website because it's not relevant to the Web3 + Full-Stack focus the hiring persona targets. Still exists on GitHub ([github/trading-strategy-simulator](https://github.com/akshansh/trading-strategy-simulator)) but isn't linked anywhere prominent.

2.5 SwapX — CEX + DEX Exchange (Full Build)

Built from scratch as a Next.js 14 app with App Router. Features:

- **CEX tab:** Order book — market orders and limit orders, 0.1% fee, per-user balances
- **DEX tab:** AMM formula — constant product swap with 0.3% fee, slippage display
- **Portfolio tab:** Holdings, allocation bars, USD values (live prices)
- **History tab:** Last 50 trades with timestamps and P&L
- **30+ tokens:** BTC, ETH, SOL, BNB, USDT, USDC, MATIC, AVAX, LINK, UNI, and more
- **Google OAuth:** Login via NextAuth.js — no wallet required
- **Redis storage:** Upstash Redis (REST API, works serverless) — portfolio as Hash, orders as List
- **Price feed:** CoinGecko free API, cached in Redis for 60s to avoid rate limits

Key files in `swapx-dex/` :

File	Purpose
<code>lib/tokens.ts</code>	30+ token definitions + INITIAL_PORTFOLIO for new users
<code>lib/redis.ts</code>	Upstash Redis client + key builders
<code>lib/auth.ts</code>	NextAuth Google OAuth config — seeds portfolio on first login
<code>lib/prices.ts</code>	CoinGecko price fetch + Redis 60s cache
<code>app/api/swap/route.ts</code>	DEX swap API
<code>app/api/order/route.ts</code>	CEX buy/sell API
<code>app/api/portfolio/route.ts</code>	Get/update portfolio
<code>app/dashboard/dashboard-client.tsx</code>	Main UI — all 4 tabs, charts, forms
<code>railway.toml</code>	Railway deployment config

2.6 Resume Website (index.html) — Major Overhaul

- Added **Summary.exe window** — at-a-glance profile with shipped projects, core stack, what I'm exploring, open to
- Updated **Skills window** with full current stack (Next.js 14, Redis, Upstash, NextAuth, Railway, CoinGecko, OAuth 2.0)
- Added **second email** `akshanshyadav26220.dpskln@gmail.com` to contact window
- Changed email to `250102005@HBTU.AC.IN` everywhere
- Added project years (2025, 2026) to all project entries
- Status widget: "OPEN TO WORK" + "SHIPPING: SwapX · FaceMetrics"
- SwapX live link added

2.7 Resume PDF (resume-print.html) — Two Redesigns

Redesign 1 (earlier session): Full visual overhaul — gradient border-top (`border-image: linear-gradient`), colored left-accent section titles, indigo pill tags for skills. Eye-catching modern layout.

Redesign 2 (this session): Reverted to clean monospace/typewriter style — Courier Prime font, no colors, plain text skill rows, dotted dividers. Removed bio/summary paragraph. All content fits in one A4 page.

2.8 GitHub Profile README (expediator/expediator)

Created the `expediator/expediator` repo (special GitHub repo that shows README on your profile). Built with:

- shields.io badges for all tech stack items
- github-readme-stats cards for GitHub stats and top languages
- streak-stats for contribution streak
- Interests, achievements, B.Tech footer
- Two email badges

Projects section removed per request — kept it focused on skills and stats.

3. Issues Faced & How They Were Fixed

Issue 1: Railway blocked deployment — CVE security vulnerabilities

Railway's security scanner rejected the deployment because `next@14.2.5` had two HIGH severity CVEs: CVE-2025-55184 and CVE-2025-67779.

Fix: Upgraded Next.js to 14.2.35

In `package.json`, changed `"next": "14.2.5" → "next": "14.2.35"` and `"eslint-config-next": "14.2.5" → "14.2.35"`. Ran `npm install`, committed, pushed. Railway then accepted the build.

Issue 2: Railway failed on startup — Redis client threw on invalid URL

During initial setup, placeholder env vars like `UPSTASH_REDIS_REST_URL=REPLACE_WITH_REAL_VALUE` caused the Upstash Redis client to throw immediately on import because the URL wasn't a valid HTTPS URL. App crashed before serving any requests.

Fix: Use a valid-format placeholder URL

Changed placeholder to `https://needed-replace-me.upstash.io` so the Redis client could initialize without throwing. Then immediately replaced with real credentials after Upstash was set up.

Issue 3: Railway "Project has no services" when setting env vars

Running `railway variables set KEY=VALUE` before the first `railway up` failed because no service existed yet in the project.

Fix: Run `railway up --detach first`

The first deploy creates the service. After that, `railway variables set` works. Always deploy first, then set env vars, then redeploy.

Issue 4: PowerShell heredoc syntax — git commit message with special characters

Using bash heredoc syntax `git commit -m "$(cat <<'EOF' ... EOF)"` fails in PowerShell 5.1 — it doesn't understand bash heredocs.

Fix: Use PowerShell here-string syntax

```
git commit -m '@'
Your commit message here.
Multi-line works fine.
 '@
```

The closing '@ **must be at column 0** (no leading spaces). Single-quoted so \$ and backticks aren't expanded.

Issue 5: Start-Process "npx" failed in PowerShell

npx is a script file (.cmd on Windows), not a Win32 executable. Start-Process "npx" throws "not a valid Win32 application."

Fix: Use cmd.exe /c npx ... or invoke directly

```
cmd.exe /c "npx railway login"
# OR just call directly in PowerShell:
npx railway login
```

Issue 6: railway service with no args hung in non-interactive mode

Railway CLI prompts interactively to select a service if no name is given. In automated scripts, this hangs indefinitely.

Fix: Always pass the service name explicitly

```
railway service "easygoing-motivation"
```

Issue 7: RNGCryptoServiceProvider — PowerShell 5.1 static method not available

[System.Security.Cryptography.RandomNumberGenerator]::GetBytes(32) is a .NET 6+ static method. PowerShell 5.1 runs on .NET Framework 4.x, which doesn't have it as a static.

Fix: Instantiate the class first

```
$rng = [System.Security.Cryptography.RNGCryptoServiceProvider]::new()
$bytes = New-Object byte[] 32
$rng.GetBytes($bytes)
[Convert]::ToBase64String($bytes)
```

Issue 8: GitHub stats card not refreshing

github-readme-stats caches stats aggressively on their Vercel deployment. Stats could appear hours old.

Fix: Add cache_seconds=1800 + include_all_commits=true

1800 seconds (30 min) is the minimum cache_seconds allowed. Also switched streak stats from the deprecated herokuapp URL to streak-stats.demolab.com which is actively maintained.

4. What You Did Manually

Manual Step 1: Google Cloud Console — OAuth Credentials

1. Go to console.cloud.google.com

2. Create project "SwapX" → APIs & Services → OAuth consent screen → External → fill name/email
3. Credentials → Create Credentials → OAuth 2.0 Client IDs → Web application
4. Add authorized redirect URI: `https://easygoing-motivation-production.up.railway.app/api/auth/callback/google`
5. Copy Client ID and Client Secret → pasted into Railway env vars

Manual Step 2: Upstash Redis Setup

1. Go to upstash.com → sign up → Create Database
2. Region: closest to you (e.g. ap-south-1 / Mumbai for India)
3. Copy REST URL and REST Token from dashboard → pasted into Railway env vars

Manual Step 3: Railway — Set Environment Variables

After `railway up` created the service, you added these 4 variables in the Railway dashboard (or CLI):

- `GOOGLE_CLIENT_ID` — from Google Cloud Console
- `GOOGLE_CLIENT_SECRET` — from Google Cloud Console
- `UPSTASH_REDIS_REST_URL` — from Upstash dashboard
- `UPSTASH_REDIS_REST_TOKEN` — from Upstash dashboard

`NEXTAUTH_URL` and `NEXTAUTH_SECRET` were set via CLI commands.

Manual Step 4: X (Twitter) Posts

Draft posts for SwapX, FaceMetrics, Space Shooter, RC Car, and the portfolio were written and provided. You copied and pasted them manually because no X API credentials were available for automated posting.

5. How to Modify Each Project

5.1 Portfolio Website — index.html

File: `C:\Users\deepi\OneDrive\Desktop\Akshansh-Resume\index.html`

What to change	Search for this in index.html
Profile bio	<code>id="win-profile"</code>
Summary window	<code>id="win-summary"</code>
Skills	<code>id="win-skills"</code>
Projects list	<code>id="win-projects"</code>
Interests	<code>id="win-interests"</code>
Contact info / emails	<code>id="win-contact"</code>
Status widget text	<code>class="widget status-widget"</code>
Music / Spotify playlist	<code>id="spotifyFrame"</code>
YouTube track list	<code>class="yt-list"</code>

⚠ CRITICAL: Cache-buster rule

Every time you edit `style.css` or `script.js`, increment `?v=N` in `index.html`:

```
<link rel="stylesheet" href="style.css?v=8">    <!-- was v=7, bump every time -->
<script src="script.js?v=8"></script>
```

Current version: **v=7**. Next edit to those files: use **v=8**.

5.2 Resume PDF — resume-print.html

File: `C:\Users\deepi\OneDrive\Desktop\Akshansh-Resume\resume-print.html`

Edit in any text editor or VS Code. Preview by opening in Chrome. After editing, regenerate the PDF:

```
$uri = "file:///C:/Users/deepi/OneDrive/Desktop/Akshansh-Resume/resume-print.html"
$out = "C:\Users\deepi\OneDrive\Desktop\Akshansh-Resume\assets\Akshansh-Yadav-Resume.pdf"
& "C:\Program Files\Google\Chrome\Application\chrome.exe" --headless --disable-gpu --no-sandbox
"--print-to-pdf=$out" "--print-to-pdf-no-header" $uri
```

5.3 SwapX — Add a New Token

File: `swapx-dex\lib\tokens.ts`

```
{ symbol: 'XYZ', name: 'TokenName', geckoId: 'coingecko-slug', color: '#ff0000', decimals: 18,
  type: 'crypto' }
```

The `geckoId` must match the CoinGecko URL slug (e.g. `bitcoin`, `solana`, `chainlink`).

5.4 SwapX — Change Trading Fees

- DEX 0.3% fee → `app/api/swap/route.ts` — change `0.003`
- CEX 0.1% fee → `app/api/order/route.ts` — change `0.001`

5.5 SwapX — Change Starting Balance for New Users

File: `swapx-dex\lib\tokens.ts` — `INITIAL_PORTFOLIO` object. Only affects new logins; existing users keep their balances.

5.6 FaceMetrics — Change Scoring Ranges

File: `face-rater\docs\index.html` — search for `IDEAL_RANGES` or the scoring function. Each metric has a min/max ideal range mapped to a 1–10 score.

5.7 Space Shooter — Tune Difficulty

File: `space-shooter\docs\index.html` — find the `// CONFIG` block near top of the `<script>` tag. Enemy spawn interval, speed, health, player speed, bullet damage, lives count — all editable there.

5.8 GitHub Profile README

File: `C:\Users\deepi\OneDrive\Desktop\repos\expediator\README.md`

Push to `expediator/expediator` repo — GitHub shows it on your profile page automatically.

6. Deployment Details

6.1 GitHub Pages (resume, face-rater, space-shooter, tic-tac-toe)

All GitHub Pages sites auto-deploy when you push to `main`. Takes ~1 minute to go live.

- **resume repo** — served from repo root
- **face-rater, space-shooter, tic-tac-toe** — served from `/docs` folder (configured in repo Settings → Pages)

```
# Push resume changes (example)
cd "C:\Users\deepi\OneDrive\Desktop\Akshansh-Resume"
git add index.html resume-print.html "assets/Akshansh-Yadav-Resume.pdf"
git commit -m "Update resume"
git push
```

6.2 Railway — SwapX

Live URL: <https://easygoing-motivation-production.up.railway.app>

Project: easygoing-motivation | **Service ID:** f9ea4e20-c582-4bd7-b0f7-da876ab42d6c

Free tier: 750 hours/month — enough for 24/7. Sleeps after 15 min idle (~30s cold start on next visit).

Region: sfo (San Francisco) — closest free region available.

To deploy a code change:

```
cd "C:\Users\deepi\OneDrive\Desktop\repos\swapx-dex"
git add .
git commit -m "your message"
git push
# Railway auto-detects push and redeploys (if GitHub is connected)
# OR trigger manually:
railway up
```

To check deployment status:

```
railway status
```

To view live logs:

```
railway logs
```

To set/update an environment variable:

```
railway variables set KEY=VALUE
```

6.3 Keep SwapX Awake (prevent Railway sleep)

Railway free tier sleeps after 15 minutes of no HTTP requests. To prevent this:

1. Go to uptimerobot.com — free account
2. Add monitor → HTTP(S) → URL: `https://easygoing-motivation-production.up.railway.app`
3. Check interval: every 5 minutes
4. Done — UptimeRobot pings it so Railway keeps it alive

7. Environment Variables Reference

SwapX — `.env.local` (for local development)

```
NEXTAUTH_URL=http://localhost:3000
NEXTAUTH_SECRET=<generate: see section 8.3>
GOOGLE_CLIENT_ID=<from Google Cloud Console>
GOOGLE_CLIENT_SECRET=<from Google Cloud Console>
UPSTASH_REDIS_REST_URL=https://your-db.upstash.io
UPSTASH_REDIS_REST_TOKEN=your-token-here
```

SwapX — Railway Production (same vars, different NEXTAUTH_URL)

```
NEXTAUTH_URL=https://easygoing-motivation-production.up.railway.app
NEXTAUTH_SECRET=KsQ74wML7MixVe5N/HzHmD1JiVrETsI5gpxz5CJuecU= ← current value
GOOGLE_CLIENT_ID=<your value>
GOOGLE_CLIENT_SECRET=<your value>
UPSTASH_REDIS_REST_URL=<your Upstash URL>
UPSTASH_REDIS_REST_TOKEN=<your Upstash token>
```

8. How to Get Each Service Credential

8.1 Google OAuth Client ID + Secret

1. Go to console.cloud.google.com
2. Select project "SwapX" (or create new)
3. APIs & Services → OAuth consent screen → External → publish
4. Credentials → Create Credentials → OAuth 2.0 Client IDs → Web application
5. Authorized redirect URIs: `https://easygoing-motivation-production.up.railway.app/api/auth/callback/google`
6. Also add `http://localhost:3000/api/auth/callback/google` for local dev
7. Copy Client ID and Client Secret

8.2 Upstash Redis URL + Token

1. Go to upstash.com → sign in with GitHub
2. Create Database → name it → pick region (ap-south-1 / Mumbai = lowest latency from India)
3. Dashboard → REST API section → copy **UPSTASH_REDIS_REST_URL** and **UPSTASH_REDIS_REST_TOKEN**

8.3 Generate NEXTAUTH_SECRET (PowerShell)

```
$rng = [System.Security.Cryptography.RNGCryptoServiceProvider]::new()  
$bytes = New-Object byte[] 32  
$rng.GetBytes($bytes)  
[Convert]::ToBase64String($bytes)  
# Copy the output – that's your secret
```

9. Quick Commands Cheatsheet

```
# Run SwapX locally
cd "C:\Users\deepi\OneDrive\Desktop\repos\swapx-dex"
npm run dev
# Opens at http://localhost:3000

# Build SwapX (test production build)
npm run build && npm start

# Push resume site changes
cd "C:\Users\deepi\OneDrive\Desktop\Akshansh-Resume"
git add index.html resume-print.html style.css script.js "assets/Akshansh-Yadav-Resume.pdf"
git commit -m "Update"
git push

# Regenerate resume PDF
$chrome = "C:\Program Files\Google\Chrome\Application\chrome.exe"
$html = "C:\Users\deepi\OneDrive\Desktop\Akshansh-Resume\resume-print.html"
$pdf = "C:\Users\deepi\OneDrive\Desktop\Akshansh-Resume\assets\Akshansh-Yadav-Resume.pdf"
& $chrome --headless --disable-gpu --no-sandbox "--print-to-pdf=$pdf" "--print-to-pdf-no-header" "file:///$(($html -replace '\\', '/'))"

# Push GitHub profile README
cd "C:\Users\deepi\OneDrive\Desktop\repos\expediator"
git add README.md
git commit -m "Update profile"
git push

# Push any project (face-rater, space-shooter, etc.)
cd "C:\Users\deepi\OneDrive\Desktop\repos\face-rater"
git add docs/index.html
git commit -m "Update"
git push

# Check Railway deployment status
railway status

# View Railway live logs
railway logs

# Set a Railway env var
railway variables set KEY="VALUE"

# Re-deploy SwapX to Railway manually
cd "C:\Users\deepi\OneDrive\Desktop\repos\swapx-dex"
railway up
```

10. GitHub Repos Summary

Repo	Live URL	Local Path	Auto-Deploy
expediator/resume	expediator.github.io/resume	Desktop\Akshansh-Resume\	GitHub Pages on push
expediator/swapx-dex	railway.app/...	Desktop\repos\swapx-dex\	Railway on push (if connected)
expediator/face-rater	expediator.github.io/face-rater	Desktop\repos\face-rater\	GitHub Pages on push
expediator/space-shooter	expediator.github.io/space-shooter	Desktop\repos\space-shooter\	GitHub Pages on push
expediator/tic-tac-toe	expediator.github.io/tic-tac-toe	Desktop\repos\tic-tac-toe\	GitHub Pages on push
expediator/gesture-robot	—	Desktop\repos\gesture-controlled-robot\	—
expediator/expediator	GitHub profile page	Desktop\repos\expediator\	Shows on profile on push